Engineering Electromagnetics II
Homework #1 Solutions
106 points total

ECE 332 – Spring 2026

Due Tuesday, April 21, 2026

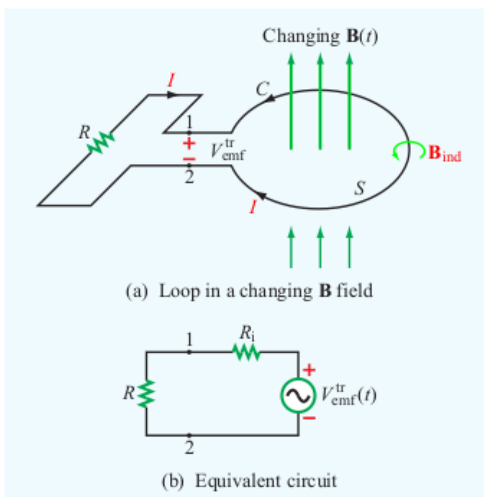


Figure 6-2 (a) Stationary circular loop in a changing magnetic field $\mathbf{B}(t)$, and (b) its equivalent circuit.

1. The loop shown above is in the x-y plane and $\mathbf{B} = \hat{\mathbf{z}} B_0 \sin \omega t$. What is the direction of I ($\hat{\boldsymbol{\phi}}$ or $-\hat{\boldsymbol{\phi}}$) at:

   (a) $t = 0$ (5 points)

   (b) $\omega t = \pi/4$ (5 points)

   (c) $\omega t = \pi/2$ (5 points)

Explain your answers. (15 points total)

Here, the loop is stationary and the magnetic field is changing. So, the total emf is the transformer emf

$$V_{\text{emf}}^{\text{tr}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}.$$

In all cases,

$$\frac{\partial \mathbf{B}}{\partial t} = \hat{\mathbf{z}} B_0 \omega \cos \omega t.$$

Also, $d\mathbf{s} = \hat{\mathbf{z}} ds$ and

$$\frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = (\hat{\mathbf{z}} B_0 \omega \cos \omega t) \cdot (\hat{\mathbf{z}} ds) = (B_0 \omega \cos \omega t) ds.$$

Therefore,

$$V_{\text{emf}}^{\text{tr}} = -B_0 \omega \cos \omega t \int_S ds = -A B_0 \omega \cos \omega t,$$

where $A > 0$ is the area of the loop.

(a) At $t = 0$,

$$V_{\text{emf}}^{\text{tr}} = -A B_0 \omega \cos 0 = -A B_0 \omega < 0.$$

This means that the current flows from the negative terminal to the positive terminal, i.e. the direction of I is $-\hat{\boldsymbol{\phi}}$.

(b) When $\omega t = \pi/4$,

$$V_{\text{emf}}^{\text{tr}} = -A B_0 \omega \cos(\pi/4) = -(\sqrt{2}/2) A B_0 \omega < 0.$$

So, the current flows from the negative terminal to the positive terminal, i.e. the direction of I is $-\hat{\boldsymbol{\phi}}$.

(c) When $\omega t = \pi/4$,

$$V_{\text{emf}}^{\text{tr}} = -A B_0 \omega \cos(\pi/2) = 0.$$

So, there is no current flowing.

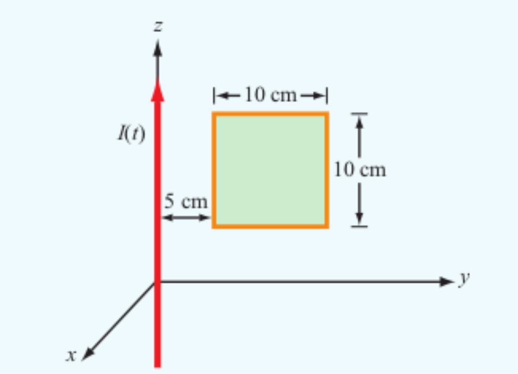


Figure P6.6 Loop coplanar with long wire (Problem 6.6).

2. The square loop shown above is coplanar with a long, straight wire carrying a current

$$I(t) = 5\cos\left(\left(2\pi \times 10^4 \text{ rad/s}\right) t\right) \text{ A}.$$

(14 points total)

(a) Determine the emf induced across a small gap created in the loop. (8 points)

The magnetic field at a distance r from a long straight wire (of length $l \gg r$) carrying a current I is

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu_0 I}{2\pi r}.$$

(See Chapter 5, Example 5-2 on pp. 237–8 in Edition 8 of the textbook.) So, the magnetic field is

$$\mathbf{B} = \hat{\boldsymbol{\phi}}\frac{\mu_0}{2\pi r}\left((5 \text{ A})\cos\omega t\right),$$

where

$$\omega = 2\pi \times 10^4 \text{ rad/s}.$$

It's possible to define the normal to the loop to be $\hat{\boldsymbol{\phi}}$ and the top and bottom edges of the loop lie along the $\hat{\mathbf{r}}$ direction and extend from $r = 5$ cm to $r = 15$ cm.

Since the loop is fixed, the emf is the transformer emf

$$V_{\text{emf}}^{\text{tr}} = -\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}.$$

In this case,

$$\frac{\partial \mathbf{B}}{\partial t} = \hat{\boldsymbol{\phi}} \frac{(5\text{ A})\mu_0}{2\pi r} \frac{\partial}{\partial t} \cos \omega t = -\hat{\boldsymbol{\phi}} \frac{(5\text{ A})\mu_0 \omega}{2\pi r} \sin \omega t$$

and

$$\frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = \left(-\hat{\boldsymbol{\phi}} \frac{(5\text{ A})\mu_0 \omega}{2\pi r} \sin \omega t\right) \cdot (\hat{\boldsymbol{\phi}} ds) = -\frac{(5\text{ A})\mu_0 \omega}{2\pi r} \sin \omega t \, ds.$$

Altogether,

$$\begin{aligned}
V_{\text{emf}}^{\text{tr}} &= -\int_S \left(-\frac{(5\text{ A})\mu_0 \omega}{2\pi r} \sin \omega t\right) ds = \frac{(5\text{ A})\mu_0 \omega}{2\pi} \sin \omega t \int_{z_0}^{z_0 + 10\text{ cm}} \int_{5\text{ cm}}^{15\text{ cm}} \frac{1}{r} \, dr dz \\
&= \frac{(5\text{ A})\mu_0 \omega}{2\pi} \sin \omega t \int_{z_0}^{z_0+10\text{ cm}} \left(\ln r|_{5\text{ cm}}^{15\text{ cm}}\right) dz \\
&= \frac{(5\text{ A})\mu_0 \omega}{2\pi} \sin \omega t \left(\ln(15\text{ cm}) - \ln(5\text{ cm})\right) \int_{z_0}^{z_0+10\text{ cm}} dz \\
&= \frac{(5\text{ A})\mu_0 \omega}{2\pi} (\sin \omega t) \ln\left(\frac{15\text{ cm}}{5\text{ cm}}\right) (z_0 + 10\text{ cm} - z_0) \\
&= \frac{((5\text{ A}) \ln 3)\mu_0 \omega}{2\pi} \sin \omega t (10\text{ cm}) \\
&= \frac{((5\text{ A}) \ln 3)(0.1\text{ m})\mu_0 \omega}{2\pi} \sin \omega t \\
&= \frac{((5\text{ A}) \ln 3)(0.1\text{ m})\left(4\pi \times 10^{-7}\text{ H/m}\right)\left(2\pi \times 10^4\text{ rad/s}\right)}{2\pi} \sin \omega t \\
&= \left(6.90 \times 10^{-3}\right) \sin\left(\left(2\pi \times 10^4\text{ rad/s}\right) t\right) \text{ AH/s} \\
&= \left(6.90 \times 10^{-3}\right) \sin\left(\left(2\pi \times 10^4\text{ rad/s}\right) t\right) \text{ V.}
\end{aligned}$$

(b) Determine the direction and magnitude of the current that would flow through a 4Ω resistor connected across the gap. The loop has an internal resistance of 1Ω. (6 points)

The current flowing across the resistor is

$$\begin{aligned}
I &= \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i} \\
&= \left(\frac{1}{4\ \Omega + 1\ \Omega}\right)\left(6.90 \times 10^{-3}\right)\sin\left(\left(2\pi \times 10^4\ \text{rad/s}\right)t\right)\ \text{V} \\
&= \left(1.38 \times 10^{-3}\right)\sin\left(\left(2\pi \times 10^4\ \text{rad/s}\right)t\right)\ \text{V}/\Omega \\
&= 1.38\sin\left(\left(2\pi \times 10^4\ \text{rad/s}\right)t\right)\ \text{mA}.
\end{aligned}$$

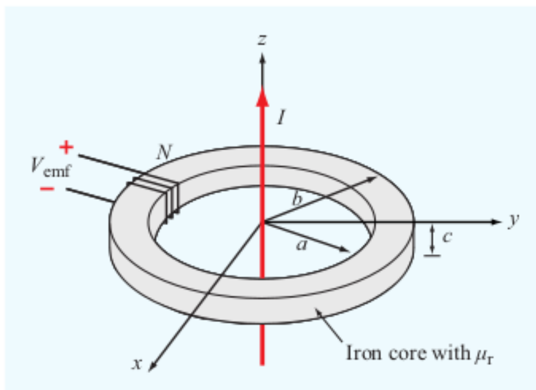


Figure P6.8 Problem 6.8.

3. The transformer shown above consists of a long wire coincident with the z axis carrying a current $I = I_0 \cos \omega t$, coupling magnetic energy to the toroidal coil situated in the $x - y$ plane and centered on the origin. The toroidal core uses iron material with relative permeability μ_r, around which 100 turns of a tightly wound coil serves to induce a voltage V_{emf}, as shown in the figure. (15 points total)

   (a) Develop an expression for V_{emf}. (10 points)

   Similar to the previous problem, the magnetic field that the wire generates is

   $$\mathbf{B} = \hat{\boldsymbol{\phi}} \frac{\mu I}{2\pi r} = \hat{\boldsymbol{\phi}} \frac{\mu I_0}{2\pi r} \cos \omega t,$$

   where μ is the permittivity of the medium, and the normal to the surface of the coil is $d\mathbf{s} = \hat{\boldsymbol{\phi}}\, ds = \hat{\boldsymbol{\phi}}\, dr dz$. Therefore, the flux through the coil is

$$\begin{aligned}
\Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} = \int_S \left(\hat{\boldsymbol{\phi}} \frac{\mu I_0}{2\pi r} \cos \omega t \right) \cdot \left(\hat{\boldsymbol{\phi}} \, ds \right) \\
&= \frac{\mu I_0}{2\pi} \cos \omega t \int_{z_0}^{z_0+c} \int_a^b \frac{1}{r} \, dr dz = \frac{\mu I_0}{2\pi} \cos \omega t \int_{z_0}^{z_0+c} \left(\ln r \big|_a^b \right) \, dz \\
&= \frac{\mu I_0}{2\pi} \cos(\omega t) \left(\ln b - \ln a \right) \int_{z_0}^{z_0+c} \, dz \\
&= \frac{\mu c I_0}{2\pi} \ln \left(\frac{b}{a} \right) \cos(\omega t).
\end{aligned}$$

So,

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = -\frac{\mu c N I_0}{2\pi} \ln \left(\frac{b}{a} \right) \frac{d}{dt} \cos(\omega t) = \frac{\mu c N I_0 \omega}{2\pi} \ln \left(\frac{b}{a} \right) \sin(\omega t).$$

(b) Calculate V_{emf} for $f = 60$ Hz, $\mu_r = 4000$, $a = 5$ cm, $b = 6$ cm, $c = 2$ cm, and $I_0 = 50$ A. (5 points)

In this case, the angular frequency is

$$\omega = 2\pi f = 120\pi \text{ rad/s},$$

the permittivity is

$$\mu = \mu_r \mu_0 = 4000 \left(4\pi \times 10^{-7} \text{ H/m} \right) = 1.6\pi \times 10^{-3} \text{ H/m},$$

the dimensions of the iron core are $a = 5$ cm $= 0.05$ m, $b = 6$ cm $=$ 0.06 m, and $c = 2$ cm $= 0.02$ m so that

$$\begin{aligned}
V_{\text{emf}} &= \frac{\mu c N I_0 \omega}{2\pi} \ln \left(\frac{b}{a} \right) \sin(\omega t) \\
&= \frac{(1.6\pi \times 10^{-3} \text{ H/m}) \, (0.02 \text{ m})(100)(50 \text{ A}) \, (120\pi \text{ rad/s})}{2\pi} \ln \left(\frac{0.06}{0.05} \right) \\
&\qquad \cdot \sin \left((120\pi \text{ rad/s}) \, t \right) \\
&= 5.50 \sin \left((120\pi \text{ rad/s}) \, t \right) \ \text{HA/s} \\
&= 5.50 \sin \left((120\pi \text{ rad/s}) \, t \right) \ \text{V}.
\end{aligned}$$

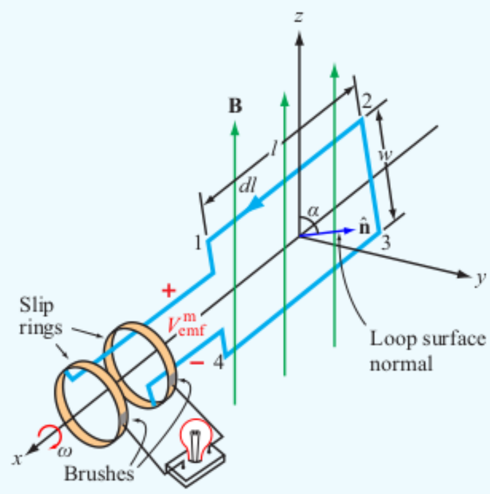


Figure 6-12 A loop rotating in a magnetic field induces an emf.

4. The electromagnetic generator shown above is connected to an electric bulb with a resistance of 150Ω. If the loop area is 0.1 m^2 and it rotates at 3,600 revolutions per minute in a uniform magnetic flux density $B_0 = 0.4$ T, determine the amplitude of the current generated in the light bulb. (6 points)

   This is exactly the type of generator analyzed in Section 6-5 of the textbook. So, the emf generated is

$$V_{\text{emf}} = A\omega B_0 \sin(\omega t + C_0),$$

   where the area of the loop is $A = 0.1$ m^2, the angular frequency is

$$\omega = (2\pi \text{ rad}) \times \left(\frac{3600 \text{ minute}^{-1}}{60 \text{ s/minute}}\right) = 120\pi \text{ rad/s},$$

   and the magnetic flux density is $B_0 = 0.4$ T. The current flowing through a bulb with resistance $R = 150\ \Omega$ is then

$$\begin{aligned} I &= \frac{V_{\text{emf}}}{R} = \frac{1}{150\ \Omega}(0.1\text{ m}^2)(120\pi \text{ rad/s})(0.4\text{ T})\sin(\omega t + C_0) \\ &= 0.10\sin(\omega t + C_0)\ \text{Tm}^2/(\Omega\text{s}) = 0.10\sin(\omega t + C_0)\ \text{V}/\Omega \\ &= 0.10\sin(\omega t + C_0)\ \text{A} \end{aligned}$$

Therefore, the amplitude of the current is 0.10 A.

5. The plates of a parallel-plate capacitor have areas of 10 cm^2 each and are separated by 2 cm. The capacitor is filled with a dielectric material with $\epsilon = 4\epsilon_0$, and the voltage across it is given by $V(t) = 30\cos\left((2\pi \times 10^6 \text{ rad/s})\, t\right)$ V. Find the displacement current. (6 points)

   This is the situation analyzed is Section 6-7 of the textbook. In this case, the area of the plates is

$$A = 10 \text{ cm}^2 = 10\left(10^{-2} \text{ m}\right)^2 = 10^{-3} \text{ m}^2,$$

   the plates are separated by a distance

$$d = 2 \text{ cm} = 2 \times 10^{-2} \text{ m},$$

   the permittivity is

$$\epsilon = 4\epsilon_0 \approx \frac{4}{36\pi} \times 10^{-9} \text{ F/m} = \frac{1}{9\pi} \times 10^{-9} \text{ F/m},$$

   the amplitude of the voltage is

$$V_0 = 30 \text{ V},$$

   the angular frequency is

$$\omega = 2\pi \times 10^6 \text{ rad/s},$$

   and the displacement current is

$$\begin{aligned}
I_\text{d} &= -\frac{\epsilon A}{d} V_0 \omega \sin \omega t \\
&= -\frac{\left(\frac{1}{9\pi} \times 10^{-9} \text{ F/m}\right)\left(10^{-3} \text{ m}^2\right)}{\left(2 \times 10^{-2} \text{ m}\right)} (30 \text{ V}) \left(2\pi \times 10^6 \text{ rad/s}\right) \sin \omega t \\
&= -\frac{1}{3} \times 10^{-3} \sin \omega t \text{ VF/s} = \frac{1}{3} \times 10^{-3} \sin \omega t \text{ V}/\Omega \\
&= -0.33 \sin\left(\left(2\pi \times 10^6 \text{ rad/s}\right) t\right) \text{ mA.}
\end{aligned}$$

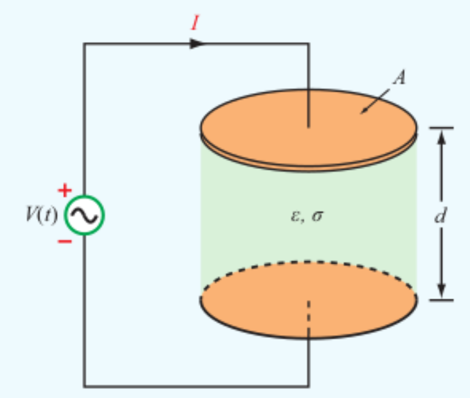


Figure P6.16 Parallel-plate capacitor containing a lossy dielectric material (Problem 6.16).

6. The parallel-plate capacitor shown above is filled with a lossy dielectric material of relative permittivity ϵ_r and conductivity σ. The separation between the plates is d and each plate is of area A. The capacitor is connected to a time-varying voltage source $V(t)$. (24 points total)

   (a) Obtain an expression for I_c, the conduction current flowing between the plates inside the capacitor, in terms of the given quantities. (6 points)

   Using the results from Example 4-14 on pp. 211-2, the conduction current is

   $$I_c = JA = \sigma EA = \frac{\sigma VA}{d},$$

   where J is the current density and E is the electric field in the dielectric.

   (b) Obtain an expression for I_d, the displacement current flowing inside the capacitor. (6 points)

   The displacement current is

   $$I_\mathrm{d} = J_\mathrm{d} A = \epsilon A \frac{\partial E}{\partial t} = \epsilon_r \epsilon_0 A \frac{\partial}{\partial t} \frac{V}{d} = \frac{\epsilon_r \epsilon_0 A}{d} \frac{\partial V}{\partial t}.$$

   (c) Based on your expressions for parts (a) and (b), give an equivalent-circuit representation for the capacitor. (6 points)

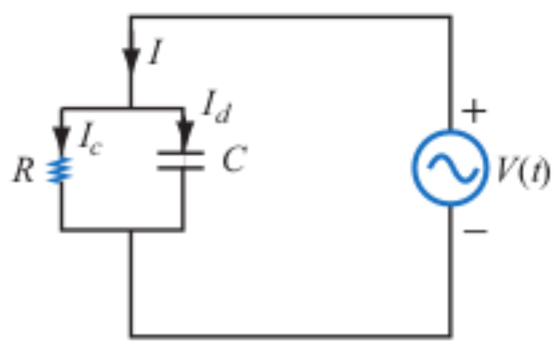
Since $I_c \propto V$ and $I_\mathrm{d} \propto \partial V/\partial t$, the conduction current acts like a current through a resistor with resistance

$$R = \frac{d}{\sigma A}$$

and the displacement current acts like the current through a capacitor with capacitance

$$C = \frac{\epsilon A}{d} = \frac{\epsilon_r \epsilon_0 A}{d}.$$

Also, since the total current $I = I_c + I_\mathrm{d}$, these components are connected in parallel in the equivalent circuit and the equivalent circuit is shown below.



Equivalent circuit

(d) Evaluate the values of the circuit elements for $A = 4\ \mathrm{cm}^2$, $d = 0.5\ \mathrm{cm}$, $\epsilon_r = 4$, $\sigma = 2.5\ \mathrm{S/m}$, and

$$V(t) = 10\cos\left(\left(3\pi \times 10^3\ \mathrm{rad/s}\right) t\right).$$

(6 points)

Here,

$$R = \frac{d}{\sigma A} = \frac{0.005\ \mathrm{m}}{(2.5\ \mathrm{S/m})\left(4 \times (10^{-2}\ \mathrm{m})^2\right)} = \frac{0.005\ \mathrm{m}}{(2.5\ \mathrm{S/m})\,(4 \times 10^{-4}\ \mathrm{m}^2)}$$
$$= \frac{50}{10}\ \mathrm{S}^{-1} = 5\ \Omega$$

and

$$\begin{aligned}C &= \frac{\epsilon_r \epsilon_0 A}{d} = \frac{4\left(8.85 \times 10^{-12}\ \mathrm{F/m}\right)\left(4 \times 10^{-4}\ \mathrm{m}^2\right)}{0.005\ \mathrm{m}}\\ &= 2.83 \times 10^{-12}\ \mathrm{F}.\end{aligned}$$

7. At $t = 0$, charge density ρ_{v0} was introduced into the interior of a material with a relative permittivity $\epsilon_r = 9$, If at $t = 1\mu$s the charge density has dissipated down to $10^{-3}\rho_{v0}$, what is the conductivity of the material? (6 points)

   The charge density at time t is

$$\rho_v(t) = \rho_{v0} e^{-(\sigma/\epsilon)t} = \rho_{v0} e^{-(\sigma/(9\epsilon_0))t}$$

   and the charge density at time $t = 1\ \mu\mathrm{s} = 10^{-6}$ s is

$$\rho_{v0} e^{-(\sigma/(9\epsilon_0))\left(10^{-6}\ \mathrm{s}\right)} = \rho_v\left(10^{-6}\ \mathrm{s}\right) = 10^{-3}\rho_{v0}.$$

   So,

$$-\frac{\sigma}{9\epsilon_0}\left(10^{-6}\ \mathrm{s}\right) = \ln 10^{-3},$$

   which means

$$\begin{aligned}\sigma &= -\frac{\left(\ln 10^{-3}\right) 9\epsilon_0}{10^{-6}\ \mathrm{s}} = -\frac{\left(\ln 10^{-3}\right) 9\left(8.854 \times 10^{-12}\ \mathrm{F/m}\right)}{10^{-6}\ \mathrm{s}}\\ &= 5.50 \times 10^{-4}\ \mathrm{S/m}.\end{aligned}$$

8. If we were to characterize how good a material is an an insulator by its resistance to dissipating charge, which of the following two materials is the better insulator? (6 points)

Dry Soil:	$\epsilon_r = 2.5$,	$\sigma = 10^{-4}$ S/m
Fresh Water:	$\epsilon_r = 80$,	$\sigma = 10^{-3}$ S/m

   The relaxation time constant for dry soil is

$$\tau_r = \frac{\epsilon}{\sigma} = \frac{2.5\epsilon_0}{10^{-4}\ \mathrm{S/m}} = \frac{2.5\left(8.854 \times 10^{-12}\ \mathrm{F/m}\right)}{10^{-4}\ \mathrm{S/m}} = 2.21 \times 10^{-7}\ \mathrm{s}$$

and the relaxation time constant for fresh water is

$$\tau_r = \frac{\epsilon}{\sigma} = \frac{80\epsilon_0}{10^{-3}\ \mathrm{S/m}} = \frac{80\,(8.854 \times 10^{-12}\ \mathrm{F/m})}{10^{-3}\ \mathrm{S/m}} = 7.08 \times 10^{-7}\ \mathrm{s}.$$

Since the time constant for fresh water is greater than the time constant for dry soil, fresh water is a better insulator than dry soil.

9. The electric field of an electromagnetic wave propagating in air is

$$\begin{aligned}\mathbf{E}(z,t) = \Big(4\cos\Big(\Big(6\times 10^{8}\ \mathrm{rad/s}\Big)\,t - (2\ \mathrm{m}^{-1})z\Big)\,\hat{\mathbf{x}} \\ + 3\sin\Big(\Big(6\times 10^{8}\ \mathrm{rad/s}\Big)\,t - (2\ \mathrm{m}^{-1})z\Big)\,\hat{\mathbf{y}}\Big)\ \ \mathrm{V/m}.\end{aligned}$$

Find the associated magnetic field $\mathbf{H}(z,t)$. (6 points)

The first step is to find the phasor $\tilde{\mathbf{E}}(z)$ of the electric field intensity. The angular frequency of the electric field intensity $\mathbf{E}(z,t)$ is

$$\omega = 6 \times 10^{8}\ \mathrm{rad/s}$$

and the wavenumber is

$$k = 2\ \mathrm{m}^{-1}.$$

So,

$$\begin{aligned}\mathbf{E}(z,t) &= (4\cos(\omega t - kz)\,\hat{\mathbf{x}} + 3\sin(\omega t - kz)\,\hat{\mathbf{y}})\ \ \mathrm{V/m} \\ &= (4\cos(\omega t - kz)\,\hat{\mathbf{x}} + 3\cos(\omega t - kz - \pi/2)\,\hat{\mathbf{y}})\ \ \mathrm{V/m} \\ &= \mathrm{Re}\Big\{4e^{j(\omega t - kz)}\hat{\mathbf{x}} + 3e^{j(\omega t - kz - \pi/2)}\hat{\mathbf{y}}\Big\}\ \ \mathrm{V/m} \\ &= \mathrm{Re}\Big\{\Big(4\hat{\mathbf{x}} + 3e^{-j\pi/2}\hat{\mathbf{y}}\Big)\,e^{j(\omega t - kz)}\Big\}\ \ \mathrm{V/m}\end{aligned}$$

Here,

$$e^{-j\pi/2} = \cos(-\pi/2) + j\sin(-\pi/2) = -j.$$

Therefore, the electric field intensity is

$$\mathbf{E}(z,t) = \mathrm{Re}\Big\{(4\hat{\mathbf{x}} - j3\hat{\mathbf{y}})\,e^{-jkz}e^{j\omega t}\Big\}\ \ \mathrm{V/m}$$

and the phasor of the electric field intensity is

$$\tilde{\mathbf{E}}(z) = (4\hat{\mathbf{x}} - j3\hat{\mathbf{y}})\,e^{-jkz}.$$

The phasor form of Faraday's law implies

$$
\begin{aligned}
\tilde{\mathbf{H}}(z) = -\frac{1}{j\omega\mu}\nabla\times\tilde{\mathbf{E}} &= -\frac{1}{j\omega\mu}\begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 4e^{-jkz} & -j3e^{-jkz} & 0 \end{vmatrix} \\
&= -\frac{1}{j\omega\mu}\left(\hat{\mathbf{x}}j3\frac{\partial}{\partial z}e^{-jkz} - \hat{\mathbf{y}}\left(-4\frac{\partial}{\partial z}e^{-jkz}\right)\right) \\
&= -\frac{1}{j\omega\mu}\left(\hat{\mathbf{x}}j3\left(-jke^{-jkz}\right) + \hat{\mathbf{y}}4\left(-jke^{-jkz}\right)\right) \\
&= \frac{k}{\omega\mu}\left(3j\hat{\mathbf{x}} + 4\hat{\mathbf{y}}\right)e^{-jkz}
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
\mathbf{H}(z,t) &= \mathrm{Re}\left\{\tilde{\mathbf{H}}(z)e^{j\omega t}\right\} = \mathrm{Re}\left\{\frac{k}{\omega\mu}\left(3j\hat{\mathbf{x}} + 4\hat{\mathbf{y}}\right)e^{-jkz}e^{j\omega t}\right\} \\
&= \frac{k}{\omega\mu}\mathrm{Re}\left\{3je^{j(\omega t-kz)}\hat{\mathbf{x}} + 4e^{j(\omega t-kz)}\hat{\mathbf{y}}\right\} \\
&= \frac{k}{\omega\mu}\left((-3\ \mathrm{V/m})\sin(\omega t-kz)\hat{\mathbf{x}} + (4\ \mathrm{V/m})\cos(\omega t-kz)\hat{\mathbf{y}}\right) \\
&= \frac{2\ \mathrm{m}^{-1}}{\left(6\times10^{8}\ \mathrm{rad/s}\right)\left(4\pi\times10^{-7}\ \mathrm{H/m}\right)} \\
&\qquad \cdot\left((-3\ \mathrm{V/m})\sin(\omega t-kz)\hat{\mathbf{x}} + (4\ \mathrm{V/m})\cos(\omega t-kz)\hat{\mathbf{y}}\right) \\
&= \Big(-0.0080\sin\left(\left(6\times10^{8}\ \mathrm{rad/s}\right)t - \left(2\ \mathrm{m}^{-1}\right)z\right)\hat{\mathbf{x}} \\
&\qquad +0.0106\cos\left(\left(6\times10^{8}\ \mathrm{rad/s}\right)t - \left(2\ \mathrm{m}^{-1}\right)z\right)\hat{\mathbf{y}}\Big)\ \ \mathrm{Vs/(Hm)} \\
&= \Big(-8.0\sin\left(\left(6\times10^{8}\ \mathrm{rad/s}\right)t - \left(2\ \mathrm{m}^{-1}\right)z\right)\hat{\mathbf{x}} \\
&\qquad +10.6\cos\left(\left(6\times10^{8}\ \mathrm{rad/s}\right)t - \left(2\ \mathrm{m}^{-1}\right)z\right)\hat{\mathbf{y}}\Big)\ \ \mathrm{mA/m}.
\end{aligned}
$$

10. Given an electric field

$$\mathbf{E} = E_0\sin(ay)\cos(\omega t - kz)\hat{\mathbf{x}},$$

where E_0, a, ω, and k are constants, find $\mathbf{H}$. (8 points)

The electric field has the form

$$\mathbf{E} = \mathrm{Re}\left\{E_0\sin(ay)e^{j(\omega t-kz)}\hat{\mathbf{x}}\right\} = \mathrm{Re}\left\{\left(E_0\sin(ay)e^{-jkz}\hat{\mathbf{x}}\right)e^{j\omega t}\right\}.$$

So, the phasor of the electric field is

$$\tilde{\mathbf{E}} = E_0 \sin(ay) e^{-jkz} \hat{\mathbf{x}}.$$

From the phasor form of Faraday's law,

$$\begin{aligned}
\tilde{\mathbf{H}} &= -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}} = -\frac{1}{j\omega\mu} \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ E_0 \sin(ay) e^{-jkz} & 0 & 0 \end{vmatrix} \\
&= -\frac{1}{j\omega\mu} \left(\hat{\mathbf{y}} \left(\frac{\partial}{\partial z} E_0 \sin(ay) e^{-jkz} \right) + \hat{\mathbf{z}} \left(-\frac{\partial}{\partial y} E_0 \sin(ay) e^{-jkz} \right) \right) \\
&= -\frac{E_0}{j\omega\mu} \left(\hat{\mathbf{y}} \left(\sin(ay) \left(-jk e^{-jkz} \right) \right) - \hat{\mathbf{z}} \left(e^{-jkz} \left(a \cos(ay) \right) \right) \right) \\
&= \frac{E_0}{j\omega\mu} \left(\hat{\mathbf{y}} jk \sin(ay) + \hat{\mathbf{z}} a \cos(ay) \right) e^{-jkz} \\
&= \frac{E_0}{\omega\mu} \left(\hat{\mathbf{y}} k \sin(ay) - \hat{\mathbf{z}} ja \cos(ay) \right) e^{-jkz}.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathbf{H} &= \mathrm{Re} \left\{ \tilde{\mathbf{H}} e^{j\omega t} \right\} \\
&= \frac{E_0}{\omega\mu} \mathrm{Re} \left\{ \left(\hat{\mathbf{y}} k \sin(ay) - \hat{\mathbf{z}} ja \cos(ay) \right) e^{j(\omega t - kz)} \right\} \\
&= \frac{E_0}{\omega\mu} \mathrm{Re} \left\{ \left(\hat{\mathbf{y}} k \sin(ay) - \hat{\mathbf{z}} ja \cos(ay) \right) \left(\cos(\omega t - kz) + j \sin(\omega t - kz) \right) \right\} \\
&= \frac{E_0}{\omega\mu} \left(\hat{\mathbf{y}} k \sin(ay) \cos(\omega t - kz) + \hat{\mathbf{z}} a \cos(ay) \sin(\omega t - kz) \right).
\end{aligned}$$